

Report on Traffic Sign Prediction

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1. Introduction

Traffic sign recognition is an essential component of autonomous driving systems and driver assistance technologies. The project aims to use a variety of machine learning approaches to categorise traffic sign photos into the appropriate groups. Classifiers are trained using various image-based and numerical feature sets, and their performance is assessed using test data that has not yet been viewed.

This report presents a classification pipeline that leverages deep learning for feature extraction, and ensemble learning techniques to achieve robust predictive performance. Deep features are extracted using a pre-trained ResNet50 Convolutional Neural Network (CNN) and apply a variety of machine learning classifiers including Random Forest, Extra Trees, Support Vector Machine (SVM), and XGBoost. These classifiers are further combined using a soft-voting ensemble to improve generalization. Our evaluation demonstrates that the ensemble approach achieves the highest cross-validation accuracy, validating the benefit of combining diverse models.

2. Methodology

2.1 Data Representation

Three types of features were provided for each sample:

- **Colour Histogram:** This feature set captures the distribution of colours in the image, providing important visual cues that are useful for distinguishing between sign types with distinct colour schemes.
- **HOG-PCA Features:** Histograms of Oriented Gradients (HOG) encode edge and shape information.

- **Additional Features:** These include numeric properties related to image edge density, mean RGB.

2.2 Feature Extraction

A pre-trained ResNet50 CNN model is used as a feature extractor. ResNet50, trained on ImageNet, is known for its deep residual connections that help preserve gradient flow in networks. We removed the final fully connected layers by setting `include_top=False` and applied global average pooling (`pooling='avg'`) to **obtain a fixed-length 2048-dimensional feature vector for each image**. Without the need for task-specific fine-tuning, this transfer learning technique enables us to take advantage of complex, hierarchical representations from a model trained on a sizable dataset.

2.3 Feature Scaling

Before further processing, all extracted features were standardized using `StandardScaler` from `scikit-learn`. Standardization ensures that each feature contributes equally to the learning process and is crucial for distance-based methods such as SVM and PCA.

2.4 Classifiers Training

1. **Random Forest:** A robust ensemble of decision trees, which also provide feature importance insights.
2. **Extra Trees:** An ensemble similar to Random Forest but with more randomness in feature splits, potentially improving generalization.
3. **SVM:** Effective for high-dimensional data; powerful classifier that maximizes the margin between classes
4. **XGBoost:** A gradient boosting framework with strong regularization capabilities, which typically performs well on structured data.

2.4.1 Ensemble Learning

Using a soft voting ensemble, the predictions from the four classifiers were aggregated. Soft voting selects the class with the highest average probability by averaging the predicted class probabilities from each classifier. This method usually outperforms individual models and makes use of the variety of model strengths.

2.4.2 Model Selection

To provide reliable evaluation across a range of class distributions, stratified 5-fold cross-validation was employed. Parameters like the number of estimators, maximum tree depth, kernel types, and learning rates using grid search for hyperparameter tweaking were optimised.

3. Results

After adding the ResNet50 CNN model, the accuracy shows a significant improvement. The classification performance of each model was evaluated using stratified 5-fold cross-validation accuracy as the primary metric. The results are shown below:

Model	CV Accuracy (before)	CV Accuracy (after)
Random Forest	0.7890	0.7340
Extra Trees	/	0.7555
SVM	0.7392	0.8216
XGBoost	/	0.7664
Ensemble	/	0.8291

Table 1- The table below summarizes the performance of the evaluated models using cross-validation accuracy as the primary metric

From the table, the ensemble model achieved the highest performance unsurprisingly, outperforming all individual models.

4. Discussion and Critical Analysis

4.1 Theoretical Model Properties

ResNet50's deep features, which offer semantically meaningful representations, greatly improved classification performance. Theoretically, deep CNNs that have been trained on huge datasets are known to perform well on related tasks.

PCA dimensionality reduction however, did not played a crucial role as expected. At first, PCA was applied to reduce noise due to high dimensionality (2048 features). It is expected to improve training efficiency and help prevent overfitting in some classifiers.

Principal components are found via PCA to capture the most variance in the data. In order to significantly shrink the feature space while maintaining most of the information, the amount of components needed to account for 95% of the total variance was kept. After comparing, the prediction accuracy is slightly better without adding PCA. This may because PCA is unsupervised, it does not consider class labels when reducing dimensions — it only cares about capturing variance. While a high variance does not mean a high relevance for classification, a complex model may not benefit from PCA. Models like SVM or Random Forest can already handle high-dimensional data well. Instead, utilizing PCA for a specific features like HOG is more effective.

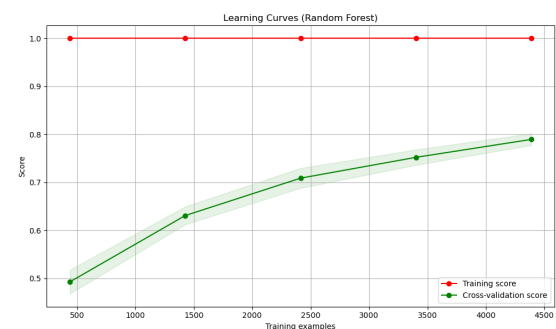


Figure 1- Learning curve for the Random Forest model

From the curve, the training score remains consistently high, indicating potential overfitting, while the cross-validation score gradually improves with more training data. The widening gap between training and validation scores suggests limited generalization, which may be mitigated by regularization or further tuning.

The Random Forest model is an ensemble-based classifier that constructs multiple decision trees during training and outputs the mode of their predictions. It is known for its robustness and ability to capture complex feature interactions, but can be prone to

overfitting, as reflected in its consistently perfect training accuracy and comparatively lower cross-validation performance. On the other hand, SVMs are particularly effective in high-dimensional spaces and tend to generalize better with proper regularization.

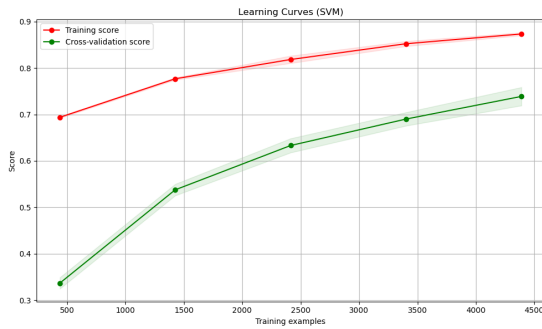


Figure 2- Learning curve of the Support Vector Machine (SVM) classifier

According to the plot, both training and cross-validation scores increase with the number of training examples. The relatively smaller gap between the two curves, compared to the Random Forest, suggests that the SVM model generalizes better and benefits more consistently from additional data. SVM, particularly with RBF kernel, showed competitive results after normalization. Its theoretical foundation lies in maximizing the margin between classes in the transformed feature space. However, its performance is sensitive to outliers and unscaled features. Without normalization, SVM performance was significantly degraded.

The learning curves suggest that while the Random Forest quickly fits the training data, the SVM improves more gradually with increasing data and demonstrates a smaller generalization gap, indicating better overall model stability and potential for improved performance with additional training samples.

The ensemble model achieved the best results due to its ability to combine the strengths of multiple learners. Soft voting improved generalization by averaging out the errors of individual models. This confirms ensemble theory: combining weak or diverse learners can lead to stronger overall performance.

One limitation is that we relied on fixed CNN features without fine-tuning the ResNet model on our task-specific data. Fine-tuning could further improve feature relevance. Additionally, while PCA helped reduce dimensions, it may also have discarded some useful variance. A more nuanced dimensionality selection strategy could be explored.

Future work could also address class imbalance using oversampling techniques like SMOTE or cost-sensitive learning. Exploring more sophisticated ensemble strategies like stacking may further enhance performance.

4.2 Feature Importance

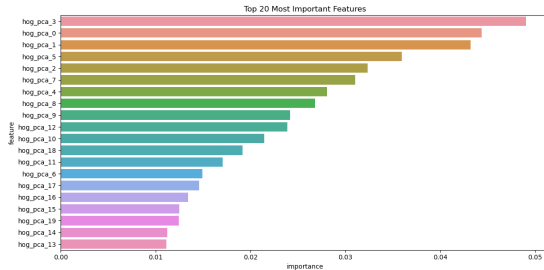


Figure 3- Top 20 most important features based on model-derived importance scores

A subset of PCA-transformed HOG features clearly contributes considerably to the model's performance, according to the feature importance study. The most important feature among the top 20 is hog_pca_3, which is closely followed by hog_pca_0, hog_pca_1, and hog_pca_5, suggesting that these elements identify important patterns in the data. The steady drop in importance values indicates that a larger group of components still contributes significantly to prediction, even though some traits are more significant. These results advocate keeping a wide variety of primary components to retain important information from the original HOG features and verify the use of PCA for dimensionality reduction.

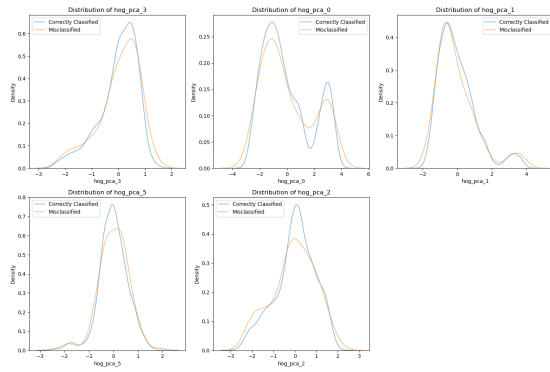


Figure 4- Kernel density plots of several HOG PCA features for both correctly identified groups and misclassified groups

Features such as hog_pca_0 and hog_pca_2 displayed noticeable shifts in density between the two groups, indicating their importance in model prediction. Meanwhile, features like hog_pca_1 showed high overlap, suggesting limited contribution to classification performance. These patterns suggest that certain HOG features may be more informative and should be prioritized in further feature selection or dimensionality reduction.

4.3 Error Analysis

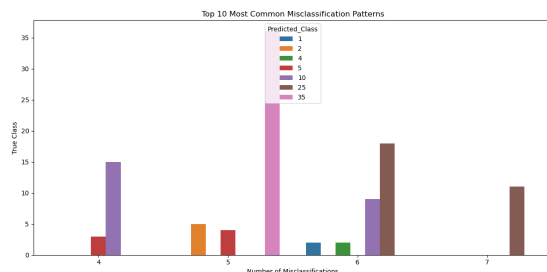


Figure 5- Visualized bar chart for understanding where the classifier is making the most errors

It is evidence from the graph that the most frequently misclassified classes are 35 and 25, suggesting that the model has difficulty distinguishing these from other classes. For future study, classes which exhibit a high misclassification counts, should be examined in detail, including feature similarity analysis.

5. Conclusions

Image classification is a fundamental problem in computer vision, with wide-ranging applications in areas like healthcare. This

project presented a machine learning pipeline for traffic sign classification using various image-derived feature representations. Deep features from ResNet50 were effective for image representation. Among the classifiers, SVM performed best individually, while the soft voting ensemble achieved the highest overall accuracy (82.9%). This study highlights the value of combining diverse methods and demonstrates the synergy between deep learning and classical machine learning techniques. Future enhancements could focus on model fine-tuning and handling data imbalance to further improve performance.

6. References

- Leo Breiman. 2001. *Random Forests*. *Machine Learning*, 45(1), 5–32.
- Vladimir N Vapnik. 1998. *Statistical Learning Theory*. Wiley.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. *Deep residual learning for image recognition*. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.
- Corinna Cortes and Vladimir Vapnik. 1995. *Support-vector networks*. *Machine Learning*, 20(3), 273–297.
- Tianqi Chen and Carlos Guestrin. 2016. *XGBoost: A scalable tree boosting system*. In *Proceedings of the 22nd ACM SIGKDD*, 785–794.
- Ian T Jolliffe and Jorge Cadima. 2016. *Principal component analysis: a review and recent developments*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065), 20150202.